

A Universe in a Bottle — Claims and Falsifiers

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2026-08-18 (revision 5; first issued 2026-06-08)

Revision 5 — 2026-08-18. One **withdrawal of a non-claim**, seven **additions**, one **correction**, and one **structural** change. **(i)** The Scope entry “ m_W and m_Z in absolute terms are not predicted” is **withdrawn**. F320 (2026-08-16) derives both absolutely; the boundary had been drawn in the wrong place, because the premise (“ v is an anchor”) was true and the conclusion did not follow from it. The masses become **core claim 12**, carrying $\rho = 1$ with them. **(ii)** Seven results issued since revision 4 are added as **claims 12-18**, with their falsifiers: the electroweak masses, the Born rule via Gleason, exact no-signalling, the finite- a Lorentz defect, the UV sector, linear structure formation, and the primordial tilt. **(iii)** The $\alpha_s(M_Z)$ row is **corrected**, not in value but in what it omitted: 0.11955 is the H1 (centre-normalised) branch of an unclosed colour-normalisation fork, and the other branch gives 0.03970. Quoting the 2.1σ without the branch overstated it. **(iv)** Revision 4 made this document a *summary* of docs/claims/ rather than the register — and then the register moved without the summary following, which is how a withdrawn claim stayed published for two days. So every claim, falsifier and scope entry below now carries a machine-checked anchor naming the card it stands on **and the status it is recording it at**. make citations fails when the two disagree, and when a headline card never reaches this page at all. The check is tools/check_summary_claims.py; the CL016 miss is its selftest fixture.

Revision 4 — 2026-08-04 - 23:55. Two changes, one an **addition** and one **structural**. **(i)** The charged-lepton **shape angle** is added as core claim 11. $\delta^* = \frac{2}{9}$ rad has been a *founding principle* of the model since 2026-07-16 (F253/F255/F256), and revision 3 — issued seventeen days later — did not carry it. That gap is the reason for the second change. **(ii)** This document is no longer the register. Every claim below now has a **card** in docs/claims/, with a status, a falsifier field, and the findings it rests on; claims-index.md lists all of them and make gate fails a card that says live while every finding under it is superseded. A prose summary could record the revision-2 withdrawals and the revision-3 correction only as blockquotes — true, but not checkable, not discoverable from the finding, and not countable. See §“Where the claims live” at the end.

Revision 3 — 2026-08-02 - 09:12. One change, and it is an **addition**, not a withdrawal: revision 2’s Scope entry on **charge quantisation was wrong**, and understated a result the tree already had. F165 (2026-06-29)

derives the hypercharge assignment up to one overall normalisation; revision 2 listed it as an input. **F279** adjudicated the contradiction by re-deriving the constraint system independently, upheld F165’s conclusion, and corrected its attribution — the closing constraint is the F47 Higgs-free Majorana step, not the gravitational anomaly. Charge quantisation moves from **Scope** to a **core claim** (new claim 10). The residual input is named: one charge unit, plus $N_c = 3$ for the specific thirds.

Revision 2 — 2026-08-02 - 10:15. Issued against audit-V items V-024, V-029, V-030 and V6.8. Four changes, all withdrawals or corrections, none an addition of new physics: **(i)** the horizon-free black hole and its four dependent falsifiers are **withdrawn** — superseded by F178, under which the exact vacuum solution is Schwarzschild and the exponential metric is the PPN-order weak-field representation only; **(ii)** the m_Z/m_W headline moves from the UV value $2/\sqrt{3}$ (1.77%) to the on-shell endpoint $3/\sqrt{7}$ (−0.064%, F49/F138), the number the tree actually computes; **(iii)** “three generations is a theorem” is corrected to match F75’s own status line — the group theory is exact, the physical identification is a stated hypothesis; **(iv)** α_s and the deuteron residuals are re-checked against PDG 2025, and a **Scope** section is added saying what is *not* claimed.

One-page summary

A deterministic **quantum cellular automaton (QCA)** on a body-centred-cubic (BCC) lattice is proposed as the physical vacuum. The free dynamics are *forced* (not chosen) by the Bisio-D’Ariano-Perinotti-Tosini uniqueness theorem: in 3D the minimal non-trivial one-particle QCA is the Weyl walk on the BCC lattice. From one reinterpretation — **the speed of light is the angular rotation rate of a real (E, B) pair per unit wavenumber**, $c_{\text{lat}} = 1/\sqrt{3}$ — the model recovers Maxwell’s equations (as a linearised rotation), the Einstein mass shell (as a spherical-Pythagorean identity), and, with gauge structure added, the Standard-Model sectors and gravity. Every structural claim is verified numerically to machine precision (residuals 10^{-14} – 10^{-16}) or as exact rationals, in a public test suite. An eleven-paper series gives the full construction.

Core claims (each a deviation from, or derivation of, the Standard Model)

1. **Light is a rotation rate, not a phase velocity** ($c_{\text{lat}} = d\Omega/d|\mathbf{k}| = 1/\sqrt{3}$). Maxwell’s curl law is its $O(\Omega)$ linearisation; energy conservation is geometric (length-preserving rotation).
2. **The photon is a bound pair of two spin- $1/2$ Weyl quanta** (de Broglie’s neutrino theory of light), not a fundamental spin-1 boson — massless, luminal, transverse, and **exactly non-birefringent**.
3. **Mass without a Higgs field:** a chiral- $SU(2)$ complex-mass step carries weak isospin as an exact gauge symmetry (Ward identity to 1.1×10^{-17}); the would-be Higgs direction is pure gauge. Hypercharge rides the same field.

4. **Exactly three fermion generations.** The group theory is a **theorem** about the cubic point group O_h : from $\sum d^2 = |O_h| = 48$ the maximal single-valued irrep dimension is 3, and the parity-odd triplet T_{1u} is unique. The **physical identification** — that a generation is that triplet, realised by the scalar mass selecting the odd-parity shell — is a **stated hypothesis**, not a theorem (F75 §7; F75 is a Candidate finding). Granting the hypothesis, a fourth generation is forbidden. Recorded this way because F79's structural G inherits the same status.
5. **The Koide relation** $Q = \frac{2}{3}$ is an exact 45° equipartition of the cubic amplitude $\sqrt{m}$ ($Q(\phi) = 1/3 \cos^2 \phi$); electromagnetism selects the colour-free charged leptons to sit there.
6. **Confinement** is exact in 2D (area law, $\sigma = -\ln w(\beta) > 0$ for all β); in 3+1D it is a colour-dielectric dual superconductor cross-checked against gauge Monte-Carlo, governed by $\mathbb{Z}_3$ centre-phase closure.
7. **The Weinberg angle is derived**, with its scale: $\sin^2 \theta_W = \frac{1}{4}$ is the **matching value at the compositeness scale** $\mu_* = 4\pi v = 3.09 \text{ TeV}$, forced by hypercharge having no lattice kinetic term (F41/F138). Running Higgs-free to M_Z gives 0.23173 (**+0.22%** vs MS-bar 0.23122), and the on-shell endpoint is $\sin^2 \theta_W^{\text{os}} = \frac{2}{9} \Leftrightarrow m_Z/m_W = 3/\sqrt{7}$ (F49). Zero new parameters; the model's two existing rulers only.
8. **Gravity is sourced by the full stress-energy tensor** — the canonical law is the induced Einstein equation $G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$ (F178). The impedance-matched lattice dielectric $K = e^{2GM/rc^2}$ (reciprocal lock $AB \equiv 1$) is its **vacuum/weak-field representation**: GR-identical PPN ($\beta = \gamma = 1$), plus the rotation-rate origin story. **Newton's constant is structural**, $G = a^2 c^3 / (8\pi \sqrt{3} \hbar)$, fixing $a/\ell_P = \sqrt{8\pi} 3^{1/4} = 6.5978$.
9. **Gravitational waves travel at exactly the photon speed**, $c_{\text{grav}} = c_{\text{lat}} = 1/\sqrt{3} = c_\gamma$, inherited through the F79 zero-tree-stiffness channel rather than imposed (F180). This is a genuine zero, not a small number.
10. **Charge quantisation is derived, not assumed** (F165, re-attributed by F279). Given the lattice-fixed representation content, the generation hypercharges are the **unique** solution — up to a single overall normalisation — of two anomaly rows, the three F27/F41 mass-step rows, and the **F47 Higgs-free Majorana step**. That last row is what closes the system: $\nu_R^\top C \nu_R$ carries hypercharge $2y_\nu$, so gauge invariance forces $y_\nu = 0$ exactly, and a Majorana mass is available only to a field of exactly zero hypercharge. The solution is $y_Q : y_u : y_d : y_L : y_e : y_\nu = 1 : 4 : -2 : -3 : -6 : 0$ with $y_\phi = 3y_Q$; normalising $y_Q = \frac{1}{6}$ gives the SM assignment and the measured electric charges $(\frac{2}{3}, -\frac{1}{3}, 0, -1)$, exactly over $\mathbb{Q}$. Both the gravitational and the cubic $U(1)^3$ anomalies are then identically zero on that line — consistency checks, not constraints. **Two inputs remain and are named**: the overall charge unit (the $\alpha/\sin^2 \theta_W$ question, F49), and $N_c = 3$ — commensurability holds for any N_c (ratios $1 : (1 + N_c) : (1 - N_c) : -N_c : -2N_c$),

so colour supplies the *value* of the fraction, not the *fact* of quantisation. Unlike the Standard-Model theorems this parallels (Minahan-Ramond-Warner; Geng-Marshak), the closing constraint here is Higgs-free structure the model already needed for the see-saw.

11. **The charged-lepton shape is fixed by a representation weight, with zero shape parameters.** The condensate angle **is** the second-shell E_g weight, $\delta^* = \dim(E_g)/\dim(T_{1u} \otimes T_{1u}) = \frac{2}{9}$ rad (exact O_h , F175) — *weight-as-phase*. δ^* is **primary** and the sextic clock coupling $\lambda_6 = 0.243$ (equivalently $W = 6\lambda_6 = 1.46$) is an **output** via the F234 arrow $\lambda_6 = |B|/(2e^6 \cos \frac{2}{3})$, not a fit. From $\{\delta^* = \frac{2}{9}, \eta^2 = \frac{1}{2}\}$ the whole charged-lepton shape follows to $\leq 0.007\%$. It is adopted because **every alternative is closed, not because it fits best**: the angle is a genuine radian with $R = 1$ forced by Schur-isotropy of the E_g irrep metric (F255, derived not posited); a scale-free topological origin is excluded, the only available holonomy being $2\pi/3$ (F253); and the dynamical Landau route provably cannot give exact $3\delta^* = Q$, since the sea B and the induced C have independent $O(1)$ origins and the identity holds only to 1.7×10^{-5} (F256). Together with claim 5 this fixes the *shape*; the overall *scale* remains an input (see Scope). Supersedes the F179/CN3 reading, under which the spectrum was a one-angle **fit**.
12. **m_W and m_Z are predicted absolutely, from two electroweak inputs where the Standard Model needs three.** Given $\{\alpha, G_F\}$ and nothing else, $m_W = \frac{3v}{2}\sqrt{2\pi\alpha}$ and $m_Z = \frac{9v}{2}\sqrt{2\pi\alpha/7}$ with $v = (\sqrt{2}G_F)^{-1/2}$ — the rationals being the BCC Wigner-Seitz facet count (7) and sublattice count (2), and the stiffness quantum $u = 18\pi\alpha/7$ being *determined* by $e = g \sin \theta_W$ rather than free. The Standard Model's third input is a measured boson mass; here $\sin^2 \theta_W^{\text{os}} = \frac{2}{9}$ is lattice geometry, so the input count falls by one and **both masses become outputs**. The residual is free of radiative corrections: model and SM sit on the same on-shell relation with the same Δr , which cancels in the ratio (verified to 2.2×10^{-16} over $\Delta r \in [0, 0.10]$), giving $+0.222\%$ on m_W and $+0.158\%$ on m_Z ; the brackets $[80.147, 80.548]$ and $[90.879, 91.332]$ GeV both contain the PDG values, and their width is the SM's own Δr_{rem} . Carried with them: $\rho \equiv m_W^2/(m_Z^2 \cos^2 \theta_W) = 1$ **exactly**, from the **rank** of the Higgs-free breaking — F41 absorbs exactly one Stueckelberg direction, so $\det M^2 \equiv 0$ identically in u , not at a tuned point — and *not* from a custodial $SU(2)$, which a Higgs-free model cannot invoke. Two inputs is not zero inputs: v remains an anchor, α remains the EM input, and the absolute masses are conditional on F141's equal-stiffness hypothesis. **This withdraws the revision-2 Scope entry** that said the absolute masses were not predicted.
13. **The Born rule is derived rather than assumed, because the rule supplies both premises of Gleason's theorem.** In ordinary quantum mechanics Gleason's theorem is not used as a derivation, since both of its premises — $\dim \mathcal{H} \geq 3$ and a non-contextual weight — are free assumptions there. In this model neither is: a measurement is not defined without a *record*, a record lives on environment cells, so the space cannot be two-dimensional; and the record channel is minimal coupling, a fixed operator carrying no reference to the measured basis. The same minimal coupling **einselects** the pointer observable instead of leaving it a

choice — $[H_{\text{int}}, \hat{n}(y)] = 0$ exactly (a literal 0.0, not 10^{-16}), so classicality is position definiteness, while spin is *not* einselected (commutator 6.2219) and a spin superposition survives until amplified into a position difference. Recorded as **contingent**: the theorem is proved on the model's own evidence, but the identification of a measurement with the model's record channel is a stated reading.

14. **No-signalling is exact, not asymptotic.** A local operation on region A leaves the reduced state at B unchanged to 7.8×10^{-16} across product, Bell/GHZ and generic entangled states and twelve local unitaries, while a unitary applied *across* the cut moves it by 0.461. Where a generic Lieb-Robinson system bounds leakage by an exponential tail, the automaton's causal cone is **strictly finite** — outside it the perturbed and unperturbed reduced states are bit-for-bit identical — and the cone is measured *tight*, exactly one site per brick-wall layer ($r \leq t$), pinned by a control that goes red when the radius is shrunk by one cell.
15. **At finite lattice spacing the entire failure of Poincaré covariance is the gradient of one scalar.** With $\Phi = (\Omega^2 - c_{\text{lat}}^2 |\mathbf{k}|^2)/2c_{\text{lat}}^2$, the whole algebra defect is $D_i = \partial_i \Phi$ — so finite- a boost covariance is *exactly* the statement that the off-shell invariant mass does not run with momentum. There is no second obstruction: $[K_i, P_j]$ has no defect at all, and the $[K_i, K_j]$ defect is the same D_i . It vanishes identically on the cubic axes, and is $O(|k|^3)$ for the paired-spinor photon against $O(|k|^2)$ on a chiral branch. The coefficient is fixed and parameter-free, but **the confrontation with dispersion data has not been done**, so this is not yet an observationally constrained claim.
16. **The model's UV sensitivity is exactly two operator coefficients, and both are Sakharov sectors.** On a lattice with a physical Brillouin-zone cutoff there are no divergences: the four QED counterterms are **finite** bare-to-measured reparametrisations, not subtractions of infinities. Enumerating every operator of superficial degree $D \geq 0$ in the full theory including gravity, and asking of each whether the model has a free parameter to absorb its cutoff-dependence, leaves exactly two that it does not — the cosmological constant ($D = 4$) and the Einstein-Hilbert term $1/16\pi G$ ($D = 2$). The model gets one right and one wrong, and the failure is **structural rather than a missing mechanism**: the two are the a_0 and a_1 moments of one zero-point sum, so any uniform reweighting moves both, and demanding the measured G *and* the measured ρ_{vac} together returns an 8.3 Gpc cell. The model cannot delete its zero-point sum without deleting its own G .
17. **Linear structure formation is fixed with zero free functions, where the EFT of dark energy has two.** In the linear, sub-horizon, quasi-static regime $\mu(a, k) \equiv 1$ and $\Sigma(a, k) \equiv 1$, with no scale and no time dependence, from three structurally independent sources (the Poisson collapse with coefficient exactly $4\pi G$; G structural on a rigid substrate so $\dot{G}/G \equiv 0$; the impedance match $AB \equiv 1$ forcing the linear-order slip to a literal zero). Hence the growth index is $\gamma_g = 6/11$ **exactly**. Because $\mu = 1$ is k -independent *by derivation* rather than by choice, the model has **no screening mechanism** — the standard escape from linear-scale bounds is unavailable, and there is nothing to tune to relieve the S_8 tension. σ_8 itself is not claimed: it is linear in $\sqrt{A_s}$, and A_s is a free initial

condition.

18. **The primordial tilt’s anomalous dimension is the anomalous part of the model’s own block-spin eigenvalue.** The $t = 0$ state of the rigid 3D automaton is a probability measure on 3D field configurations — a 3D Euclidean statistical field theory — so the primordial spectrum is that measure’s energy-energy correlator and $\gamma \equiv \frac{1}{2}(1 - n_s) = y - 1$ **identically**, with no holographic dual required or claimed. Two consequences follow, and the first is a bill the model has not yet paid: every block-spin eigenvalue exponent measured so far is an integer, which gives $n_s = 1$ exactly — Harrison-Zel’dovich, excluded at $8.4\text{--}9.9\sigma$ — so the model *requires* exactly one non-integer eigenvalue of size $\gamma \approx 0.014\text{--}0.018$, and producing it is the open calculation. Second, the stress tensor’s dimension is protected by its own conservation, giving $n_t = 2$ **exactly** (strongly blue) and $\log_{10} r(k_*) \approx -118$. Contingent on the statistical-field identification.

Withdrawn in revision 2. The former claim 9, “black holes are horizon-free dielectric condensates”, is **retracted**. F178 demotes the exponential metric to the PPN-order weak-field representation; the **exact vacuum solution is Schwarzschild**, with a horizon. The horizon-free throat, the $+4.63\%$ shadow, the absent Hawking spectrum, the late-time ringdown echoes and the 4π second-order deflection coefficient were all consequences of treating the exponential form as fundamental, and none of them is predicted by the model as it now stands. They are listed here, rather than deleted silently, because they were published as live falsifiers between 2026-06-08 and 2026-08-02.

Headline verified numbers

All reference values are **PDG 2025 / CODATA 2022** as of 2026-08-02.

Quantity	Model	Reference	Residual	Free parameters
c_{lat}	$1/\sqrt{3} = 0.5773503$	— (definition of the lattice unit)	exact	0
BCC lattice constant a	$2/\sqrt{3} = 1.1547005$	— (closed form, F278)	exact	0
$\sin^2 \bar{\theta}_W(M_Z)$	0.23173	0.23122	$+0.22\%$	0
m_Z/m_W (on-shell, $\sin^2 \theta_W^{\text{os}} = \frac{2}{9}$)	$3/\sqrt{7} = 1.133893$	91.1880/80.3692 = 1.134614	$\mathbf{-0.064\%}$	0
Koide Q	0.6666605	$\frac{2}{3}$	0.91σ ; predicts $m_\tau = 1776.97$ MeV (6×10^{-5})	0

Quantity	Model	Reference	Residual	Free parameters
Electric charges (Q_u, Q_d, Q_ν, Q_e)	$(\frac{2}{3}, -\frac{1}{3}, 0, -1)$	same	exact over $\mathbb{Q}$ (literal 0)	1 (charge unit) + $N_c=3$
$a/\ell_P; G$	6.5978; 6.6743×10^{-11}	CODATA	3×10^{-8}	0
m_W (absolute, from $\{\alpha, G_F\}$)	$\frac{3v}{2}\sqrt{2\pi\alpha} \in$ [80.147, 80.548] GeV	$80.3692 \pm$ 0.0133	+0.222% (Δr -free)	0 (2 inputs, not 3)
m_Z (absolute, same two inputs)	$\frac{9v}{2}\sqrt{2\pi\alpha/7} \in$ [90.879, 91.332] GeV	91.1880	+0.158% (Δr -free)	0 (2 inputs, not 3)
$\rho =$ $m_W^2/(m_Z^2 \cos^2 \theta_W)$ (not custodial $SU(2)$)	1, from rank u	1	exact over $\mathbb{Q}$, identically in u	0
Growth index $\gamma_g; \mu, \Sigma$	6/11; $\equiv 1$	GR: 6/11; $\equiv 1$	exact (no screening available)	0
$\alpha_s(M_Z)$, H1/centre branch	0.11955 (1-loop)	$0.1175 \pm$ 0.0010	+1.7% (2.1 σ)	1 scheme- matching input; branch unchosen
Deuteron E_b	2.224 MeV	2.22457 MeV	0.026%	1 ($b = 0.55$ fm)

Two residuals moved since revision 1, and in both cases the *measurement moved, not the model*. $\alpha_s(M_Z)$ was recorded at +1.3% against a PDG average of 0.1180; that average is now 0.1175 ± 0.0010 , so the same unchanged model number is +1.7%, i.e. $\approx 2.1\sigma$ of experiment. This is the register's largest open tension and is stated as such.

And that number carries a branch, named here from revision 5. 0.11955 assumes the model's bare coupling is centre-normalised, $1/(16\pi)$ — hypothesis **H1** of an unclosed colour-normalisation fork. On the other branch, **H2**, the coupling is $1/(16\pi C_F)$ and the same chain gives $\alpha_s(M_Z) = 0.03970$: not a 2.1σ tension but a -66% miss. The model's own exactly solvable 2D $SU(3)$ engine measures $\sigma_6/\sigma_3 = 2.4911511$ at its own β , against 5/2 for Casimir and 1 for centre — i.e. it measures **H2**, while the α_s chain requires H1, and no argument for the centre reading has been found (three candidate reconciliations are closed exactly). The deciding computation is named and cannot come out ambiguous: a one-loop background-field Λ -ratio for the model action, which must be ≈ 1.78 under H1 and $\approx 3.4 \times 10^6$ under H2. **Until it is done this row is one branch of a fork the tree cannot close, and quoting the 2.1σ alone overstates it.** The deuteron moved the other way, $0.34\% \rightarrow 0.026\%$ — but at a chosen $b = 0.55$ fm, so it is a one-parameter result and is no longer described as zero-parameter.

One prediction survived an out-of-sample revision. $m_Z/m_W = 3/\sqrt{7}$ was fixed before PDG 2025 excluded the CDF-II m_W measurement for low compatibility. Against the resulting $m_W = 80.3692 \pm 0.0133$ the model gives -0.064% (implied $m_W = 80.4203$). Had it been tuned to CDF it would now look worse; it was not, and it survives the exclusion cleanly.

Falsifiable predictions (with thresholds)

- **Quantum-gravity dispersion.** A quadratic ($n = 2$) vacuum dispersion with energy scale $E_{\text{QG},2} = \sqrt{54} \hbar c/a \approx 1.36 \times 10^{19}$ GeV. **Any measured $n = 2$ time-of-flight bound above 1.36×10^{19} GeV kills the adopted lattice cell.**
- **Vacuum birefringence.** The physical (paired) photon is **exactly non-birefringent** — not within a tolerance, but structurally: the pair carries one single-valued rotation rate Ω_{pair} , so there is no second branch to split from. A confirmed first-order vacuum birefringence in GRB/AGN polarimetry would contradict it (and would, conversely, revive the excluded chiral construction).
- **Gravitational-wave speed.** $c_{\text{grav}} = c_\gamma$ **exactly**, with no free coefficient (F180; GW170817 residual $\leq 3 \times 10^{-83}$). Any confirmed non-zero $c_{\text{grav}}/c_\gamma - 1$ falsifies the inheritance mechanism.
- **Weinberg angle.** The chain is committed to $\sin^2 \theta_W = \frac{1}{4}$ at $\mu_\star = 4\pi v$ and $\sin^2 \theta_W^{\text{os}} = \frac{2}{9}$ on shell. A precision m_W that moved m_Z/m_W off $3/\sqrt{7}$ by more than the running uncertainty would falsify it — which makes the PDG 2025 re-analysis (CDF-II excluded) a passed test, not a retrofit.
- **Charged-lepton shape.** A charged-lepton mass measurement inconsistent with the $\{\delta^* = \frac{2}{9}, \eta^2 = \frac{1}{2}\}$ shape at better than 0.007%. Because δ^* is an exact rational fixed by O_h representation theory, **there is no parameter to re-fit** — the two no-gos (F253, F256) closed the alternatives deliberately, and that is what makes this falsifiable rather than adjustable.
- **Absolute m_W .** The whole residual on claim 12 is the on-shell Weinberg angle and there is no parameter left to absorb a change in it: the claim dies if $\sqrt{\sin^2 \theta_W^{\text{obs}}/(2/9)}$ leaves $[0.995, 1.005]$, equivalently if the measured m_Z/m_W moves off $3/\sqrt{7}$ by more than 0.1%. A measurement of m_W at fixed $\{\alpha, G_F\}$ outside $[80.147, 80.548]$ GeV that is not attributable to Δr kills it directly.
- **No dimension-5 photon operator.** The leading Lorentz-violating photon operator is dimension-6, established by measuring the scaling exponent $p = 2$ rather than by bounding a coefficient, with the exact rational coefficient $-(1 - \sum_i \hat{n}_i^4)/144 + (\hat{n}_x \hat{n}_y \hat{n}_z)^2/24$ (verified to 1.1×10^{-20} in 60-digit arithmetic; range exactly $[-1/162, 0]$, vanishing along $\langle 100 \rangle$). **Any detection of dimension-5 photon Lorentz violation — linear energy-dependent arrival times, or birefringence — falsifies it**, because the claim is that the operator is *absent*, not small. This is also why the paired photon survives the GRB bound that excluded the chiral σ -bilinear construction.
- **S_8 and late-time growth.** Claim 17 predicts the combined-CMB S_8 propagated forward by GR growth, with nothing to tune: 0.8410 against combined CMB $0.836^{+0.012}_{-0.013}$ (0.29σ), KiDS-Legacy 2025 (1.17σ) and **DES Y6 $3 \times 2\text{pt}$ 0.789 ± 0.012**

(3.00σ). If the DES Y6 direction consolidates as physical rather than as photo- z or baryonic-feedback systematics, the claim is dead and the gravity sector with it — and the usual escape, modified gravity confined to nonlinear scales by screening, is unavailable here. Secondary thresholds: $|\mu - 1| > 0.011$, a detected linear-order slip $|\eta - 1| \gg 2 \times 10^{-5}$, or a measured $\dot{G}/G \neq 0$.

- **Tensor-to-scalar ratio.** Claim 18 gives $n_t = 2$ exactly and $r \sim 10^{-118}$ at the CMB pivot, with no parameter to move: **any detection of r at all** — BICEP Array targets $\sigma(r) \lesssim 0.003$ — falsifies it. The claim also requires exact power laws, so a significantly non-zero running $dn_s/d \ln k$ falsifies it, with the discriminator at $\sigma \sim 2.6 \times 10^{-4}$.
- **Mercury precession.** PPN $\beta = \gamma = 1$ (42.98"/cy). The naive *linear* dielectric ($\beta = \frac{1}{2}$, 50.1"/cy) is excluded. Under F178 this is now a consistency requirement rather than a distinctive prediction: in vacuum the model **is** GR, so a confirmed PPN deviation falsifies it exactly as it would falsify GR.

Withdrawn in revision 2 (all four were consequences of the pre-F178 exponential-metric-as-fundamental reading, and the model no longer predicts any of them): photon-ring imaging at +4.63%; gravitational-wave ringdown echoes; absence of a thermal Hawking spectrum; the 4π second-order light-bending coefficient. **An observation matching Schwarzschild in any of these four would previously have been recorded as falsifying the model. It does not.**

Scope — what is *not* claimed

Stated explicitly so that absence is not read as a prediction. Each is open and acknowledged, not quietly omitted.

- ~~m_W and m_Z in absolute terms.~~ **Moved to core claim 12 in revision 5.** Revision 2 recorded this as a scope boundary; F320 (2026-08-16) shows the boundary was in the wrong place. The premise was true and remains true — v is an anchor, not an output — but the absolute masses do not follow from v alone, and eliminating the stiffness quantum against $e = g \sin \theta_W$ determines them. What is still an input is named in claim 12: v , α , and F141's equal-stiffness hypothesis.
- **The CKM matrix and measured CP violation.** The construction is first-generation; $J = 0$ is the arithmetic consequence of one generation ($N_{\text{phase}} = (n-1)(n-2)/2$), not a prediction that the CKM phase vanishes (F53). Three-generation mixing is out of scope.
- **Neutrino mass scale.** F47 supplies a Higgs-free see-saw *mechanism*; nothing in the model fixes M_R , so no absolute neutrino mass is predicted.
- **Muon $g - 2$ beyond QED.** The QED piece is computed through two loops ($A_2(\mu) = 0.765857$ vs the known 0.765857410). Hadronic vacuum polarisation, hadronic light-by-light and electroweak contributions are **not** computed and not claimed.
- **Charge quantisation.** **Moved to core claim 10 in revision 3.** Revision 2 listed this as an input; that was wrong. It is derived (F165, re-derived and re-attributed by F279). What remains an input is the **one charge unit** — the overall normalisation, which is the $\alpha / \sin^2 \theta_W$ question (F49) — and, for the specific *thirds* rather than commensurability as such, $N_c = 3$. Both are named in claim

10.

- **“No doublers” has a domain.** The BCC Weyl walk has exactly one zero, at $k = 0$, and no $\omega = \pi$ point **on the cubic FFT grid** (L up to 128, both branches). On the true BCC zone the $\omega = \pi$ mode sits exactly at the corner H, $|\mathbf{k}| = 2\pi/a = \pi\sqrt{3}$ (F278). Every observable in the tree is computed on the grid, so no result depends on the distinction — but the unqualified form of the claim overstates it, and the discriminating test is not yet built.
- **The cosmological constant** is not derived; F193/F196 reduce the classic 121-order problem to the $\Omega_\Lambda \approx 0.69$ coincidence, which is not the same as explaining it. Revision 5 sharpens the *shape* of what is missing rather than the result: by claim 16 the one obvious internal route — reweighting the zero-point sum — is **excluded**, because the same sum supplies G . Whatever solves this is not a rescaling.

Where the claims live

Since 2026-08-04 this document is a **summary**, not the register. The register is docs/claims/ — one card per claim, CL{NNN}-{slug}.md, each carrying a present-tense status (live · narrowed · contingent · open · withdrawn · not_claimed), a falsifier field, the findings it rests on, and its own revision history. claims-index.md lists every card.

The distinction the cards exist to make: **a finding is a research record and a claim is a position**. A finding is written once and superseded rather than rewritten — it is correct for it to keep saying what a session concluded. The claim resting on it is what has to move, and until now there was nowhere to move it. Sixteen findings in this project are named in a superseded: list while their own headers still read Confirmed – N/N PASS; that is not a defect in the findings, it is the gap the cards fill.

Three consequences worth stating here, because they are what a reader of *this* document should know:

- **The revision-2 withdrawals are cards, not a blockquote.** The horizon-free black hole and its four dependent falsifiers are CL023–CL027, each status: withdrawn, each card’s history section *being* the retraction record. A gate test asserts they are never deleted, because a withdrawn claim that is deleted is a claim that gets re-made.
- **Overstatement is now machine-checkable.** tools/check_claims.py fails a card marked live when *every* finding it rests on is named in a superseded: list. On its first run it caught seven. It deliberately does **not** fire on partial supersessions: of fourteen files one audit called superseded, exactly one was.
- **Not every card is authored.** 58 are; 223 are review_state: unreviewed-seed — extracted from a finding, classification *not* confirmed by a reviewer, and not citable as independent support. The count is a ratcheted debt, not a claim of 223 results.
- **This document can no longer drift from the register.** Every claim, falsifier and scope entry above carries an anchor — an HTML comment, invisible in the PDF — naming the cards it stands on *and the status it records them at*, and make citations fails when a card’s status moves and this page does not follow

it. Added 2026-08-18, after CL016 stayed published here for two days past its withdrawal; that miss is the check's own selftest fixture. A headline card that never reaches this page at all is counted as declared debt rather than failing the gate, so a new result can be recorded before it has been written up for a lay reader.

Reproducibility

Every claim above runs in a public test suite to the stated residual. The **structural** sector — lattice cell, gauge couplings, metric — carries **no free fit parameters**; the two rows that do carry one (α_s , the deuteron) are marked in the table rather than folded into a headline. Source, tests, and per-finding write-ups accompany the papers.

Barrier: make gate. A green bare pytest is not sufficient — it cannot collect the three scenario records that run the engine on a lattice.

Full series (11 papers) and this summary: see the accompanying papers/ directory. Contact: benludwig6382@gmail.com.